

## LAB 4 – SECOND-ORDER FILTERS

### Introduction

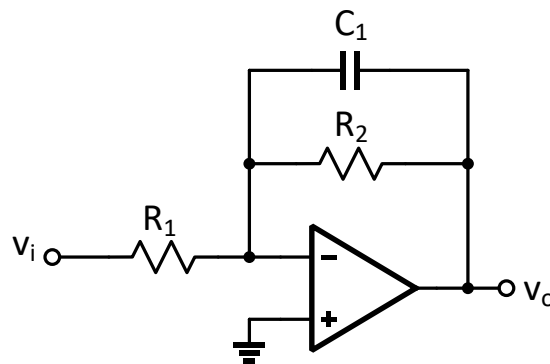
In this lab, you will look at properties of second-order filters and measure the frequency response of various second-order systems.

### Experiment 1. Low-Q Low-Pass Active Filters

This experiment focuses on realizing a second-order system by cascading two first-order systems. The systems shown above use the LMC660 op amp with  $\pm 8$  V supplies. All other component values are given in the table below.

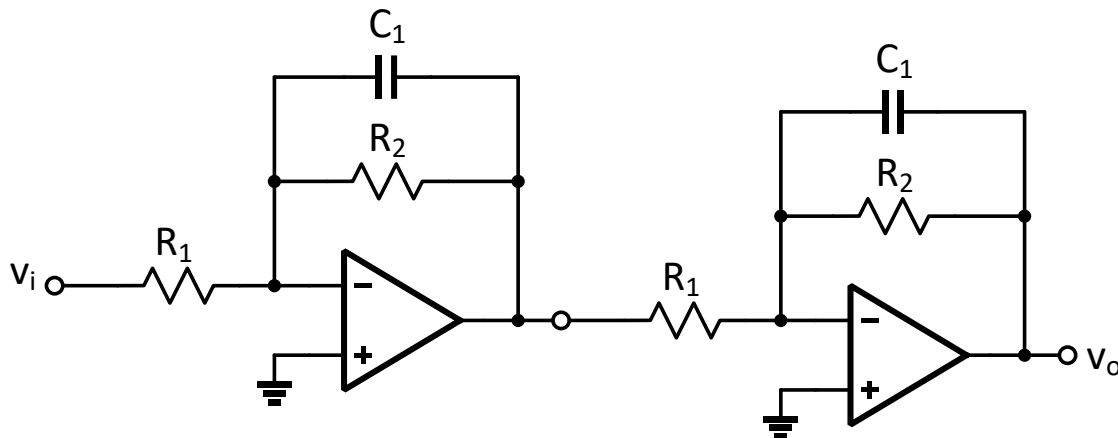
| Component | Value                |
|-----------|----------------------|
| $R_1$     | $4.7\text{ k}\Omega$ |
| $R_2$     | $10\text{ k}\Omega$  |
| $C_1$     | $10\text{ nF}$       |

We will first examine the first-order system shown below.

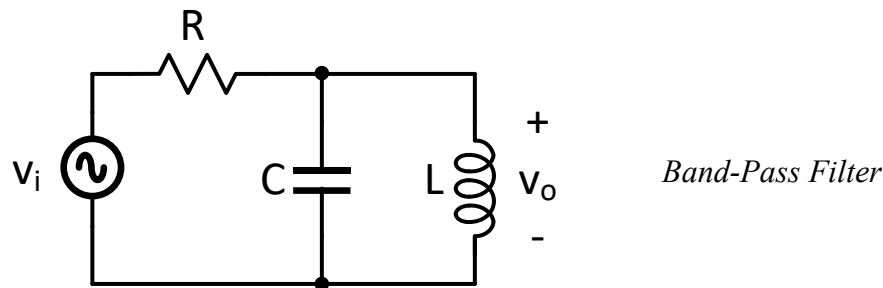
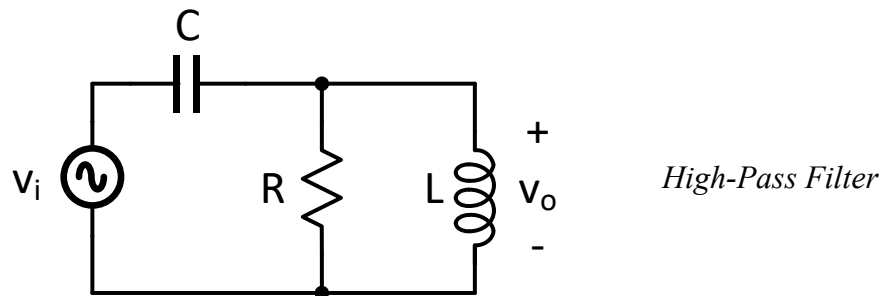
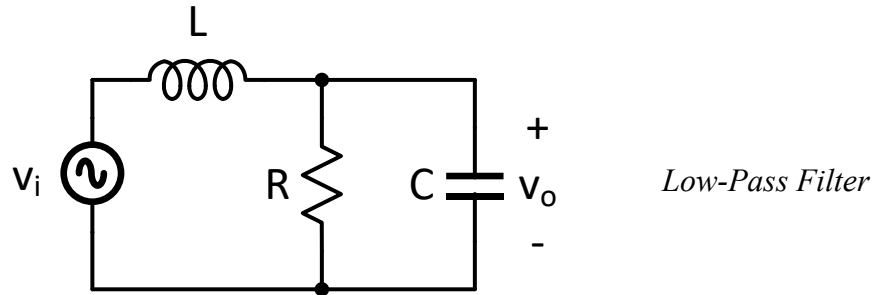


1. Compute the transfer function  $H_1(s) = v_o/v_i$  for the first order system shown above. Calculate the expected corner frequency,  $f_c$ , and low-frequency gain  $G_0$ .
2. Build the first-order system. The input signal should be a sine wave. Adjust the amplitude so that the output does not clip and set the frequency to 100 Hz. Use the Loopinator to measure the frequency response over the range of 100 Hz to 100 kHz with at least 15 points per decade.
3. Plot the frequency response obtained from the Loopinator on the same set of axes as the frequency response obtained from evaluating  $H_1(s)$ . Include this in your lab report.

We now cascade two copies of the first-order system to realize a second-order system.



4. Compute the transfer function  $H_2(s) = v_o/v_i$  for the second order system and calculate the expected values for  $f_0$ ,  $Q$ , and  $G_0$ . Notice that  $H_2(s)$  can be obtained from  $H_2(s) = H_1(s) \cdot H_1(s)$ .
5. Build the second-order circuit. Again, adjust the amplitude of the input so that the output does not clip. Use the Loopinator to measure the frequency response over the range of 100 Hz to 100 kHz with at least 15 points per decade.
6. Plot the frequency response obtained from the Loopinator on the same set of axes as the frequency response obtained from evaluating  $H_2(s)$ . Include this in your lab report.

**Experiment 2. High-Q Second-Order RLC Filters**

This experiment looks at creating second-order systems from passive RLC circuits. All three circuits shown above use the same components, but with different configurations. The component values are shown in the table below.

| Component | Value                            |
|-----------|----------------------------------|
| $R$       | $150\ \Omega$ and $1.5\ k\Omega$ |
| $L$       | $15\ mH$                         |
| $C$       | $0.1\ \mu F$                     |

Repeat the following steps for all three circuits.

1. Calculate  $H(s) = v_o/v_i$  with  $R = 1.5\ k\Omega$ , and find the expected values for  $f_0$ ,  $Q$ , and  $G_0$ .
2. Use the Loopinator to measure the frequency response for each circuit from 100 Hz to 100 kHz with at least 15 point per decade. Perform this measurement twice, once with  $R = 150\ \Omega$  and again with  $R = 1.5\ k\Omega$ .

3. Plot the measured magnitude responses on the same plot. Include this in your lab report.
4. Plot the measured phase responses on the same plot. Include this in your lab report.

Real inductors always have some series resistance due to the resistivity of the wire in the coil. For the 15-mH inductors in your lab kit, the series resistance is approximately  $10\ \Omega$ . This may have a noticeable effect on the measurements of the above circuits. For example, it may make the bandpass frequency response curve asymmetric, or it may result in additional loss in the filter. If you see these effects in your measurements, you should make note of them and discuss them in your report.

### Experiment 3. Active Filter Design

Design, build and test a unit-gain, low-pass Sallen-Key filter. Use the LMC660 op amp with  $\pm 8$  V supplies along with resistors and capacitors from you lab kit. Your filter should meet the following specifications:

| Parameter                | Value                   |
|--------------------------|-------------------------|
| Corner Frequency (kHz)   | 1 kHz $\pm 10\%$        |
| Quality Factor           | $1/\sqrt{2}$ $\pm 10\%$ |
| Low-Frequency Gain (V/V) | 1 $\pm 5\%$             |

1. Include the schematic of your design along with all derivations in the lab report.
2. Use LTSpice to perform AC simulations of your proposed design from 100 Hz to 100 kHz. Include a screenshot of the simulation results in your lab report.
3. Use the Loopinator to measure the frequency response from 100 Hz to 100 kHz with at least 15 points per decade. Include a plot of the measured frequency response (magnitude and phase) in your lab report.